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Contents

List of Figures

ABSTRACT

Wireless communication appears in almost every electronic device today, including Wi-Fi routers, Bluetooth gadgets, smart home systems, and IoT devices. All these systems require an antenna to send and receive signals. In this project, a circular microstrip patch antenna is designed for the 2.45 GHz frequency, which is commonly used for Wi-Fi, Bluetooth, and other short-range wireless applications. The antenna uses an FR4 substrate, which is low-cost and easy to make in college labs. The design started with calculations using standard formulas. Then, it was modeled and simulated using ANSYS HFSS, a software tool for studying antenna behavior. The simulation results reveal important performance parameters such as return loss (S11), VSWR, radiation pattern, and gain. The antenna resonates at 2.45 GHz with good matching and acceptable radiation performance. This project aids in understanding how to design, analyze, and optimize antennas using simulation tools. The results indicate that the circular microstrip patch antenna is suitable for simple wireless applications and can be improved in the future by increasing bandwidth, boosting gain, or enabling operation at multiple frequencies.

Keywords:Microstrip Patch Antenna, Circular Patch, 5G, IoT, Sub-6 GHz, Axial Ratio,VSWR.

Chapter 1

INTRODUCTION

1 INTRODUCTION

1.1 Introduction

Wireless communication has become an essential part of modern technological development, enabling information exchange between systems, devices, and users without the need for physical wired connections. Applications ranging from smart appliances used in the home to advanced military navigation systems rely on wireless networks for efficiency, reliability, and mobility. The antenna is at the heart of each and every system involving wireless communication; it serves as the interfacing element for the conversion of electrical signals into electromagnetic waves and vice-versa.

With the continuous development of electronic equipment toward compactness, portability, and high integration, highly demanding needs have been placed on antennas, which are required to be small in size, lightweight, low-profile, and compatible with PCB fabrication. Among various antenna types, microstrip patch antennas have become one of the most popular choices owing to their planar profile, low expense, straightforward geometry, and easy integration with RF circuits.

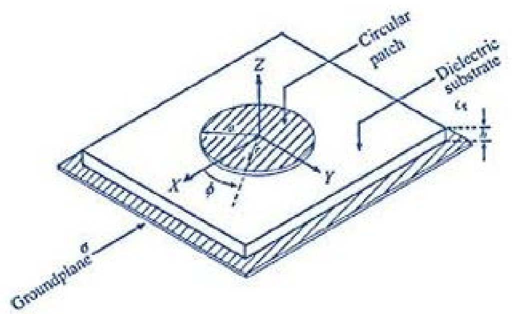


Figure 1.1: Basic Structure of The Proposed Antenna

A microstrip patch antenna essentially consists of a metallic radiating patch etched on a dielectric substrate with a ground plane on the opposite side. Among many patch geometries, the circular microstrip patch antenna is highly important owing to its rotational symmetry, stable radiation characteristics, and ability to support various polarization types. The circular patch geometry is highly suitable for wireless systems where compactness, uniform radiation, and ease of fabrication are considered important.

A circular microstrip patch antenna that operates at 2.45 GHz, falling within the globally accepted ISM band, is to be designed and analyzed in this work. This band is used in many applications, including Wi-Fi (IEEE 802.11b/g), Bluetooth, ZigBee, RFID, wireless sensor networks, and IoT platforms.

FR4 substrate (r = 4.4) was used in the design of the antenna due to its inexpensiveness, easy availability, mechanical stability, and compatibility with usual methods of PCB manufacturing. Although FR4 substrate shows higher dielectric losses compared to specialized RF substrates, such as Rogers or Taconic materials, it remains practically the most suitable choice in an academic project or prototype development.

The antenna will be modeled and simulated using ANSYS HFSS, the high-precision electromagnetic simulation tool based on the Finite Element Method. HFSS will enable full-wave 3D field analysis that gives an accurate evaluation of antenna parameters such as return loss (S11), Voltage Standing Wave Ratio (VSWR), bandwidth, impedance matching, radiation patterns, directivity, gain, and surface current distributions.
The design, implementation, and analysis of this antenna provide valuable insight into RF engineering, electromagnetic theory, and the design of wireless communication systems. The knowledge acquired through this work is useful in a number of career options: embedded systems, antenna design, satellite communication, development of smart devices, radar systems, and IoT technology..

Voltage Standing Wave Ratio VSWR represents the quality of impedance match. The ideal value is close to 1, while for practical cases, it can be up to 2. Bandwidth BW is the range of frequencies where S¡-10 dB, thus providing stable performance to the antenna. Gain and directivity represent the capability of the antenna to focus radiated energy in a particular direction. Radiation efficiency is the ratio of the power radiated to the input power, accounting for the loss brought in by dielectric and conductor.

Axial ratio AR is one of the primary parameters for the circular polarization; an axial ratio below 3 dB confirms the good characteristics of the designed antenna for CP. The radiation pattern gives a pictorial representation showing how power is radiated in space. It reflects the directional behavior of the antenna. Input impedance Z should be 50 to effectively match with standard RF components.

The relevance of antenna design continues to grow as emerging technologies demand more efficient and compact wireless communication. Smart wearable devices, health monitoring patches, autonomous robots, remote sensing systems, drones, satellite-assisted IoT. Knowledge gained from this project lays a foundation for future specialization in advanced fields such as MIMO systems, phased array antennas, 5G communication hardware, radar antenna systems, adaptive beam-forming, and millimeter-wave technologies.

1.2 Problem Statement

Achieving proper impedance matching, acceptable return loss, and stable radiation characteristics remains a significant challenge when designing low-cost antenna solutions for wireless systems operating at 2.45 GHz. The use of FR4 as a substrate further contributes to performance limitations due to dielectric losses and bandwidth reduction. Therefore, this project focuses on addressing these constraints by designing and simulating a circular microstrip patch antenna using ANSYS HFSS to obtain improved impedance matching and reliable radiation performance suitable for practical wireless communication applications..

1.3 Objectives of the Project

1. Designing and simulating circular microstrip patch antenna.
2. To simulate the antenna in HFSS and evaluate its S11, VSWR, radiation pattern, and gain for stable wireless performance.
3. Enhance antenna performance with slots and feeding methods.

1.4 Methodology/Block Diagram

The objective of this research project is to design and analyze, using CST/HFSS, a circular microstrip patch antenna . This involves antenna performances analysis such as radiation pattern, gain, axial ratio, and return-loss characterization. Testing of prototypes, the effects of the real environment, and actual fabrication are not part of this study. Some assumptions in this study include ideal material properties and free-space conditions without interference. One of the limitations of this project includes the fact that the work has been pursued based solely on simulation results without any experimental verification. Moreover, the number of design iterations to be performed has been limited by time and computational resources.

1.5 Tools Required

HFSS

1.6 Advantages and Disadvantages

Advantages

1. Low Profile and Lightweight

   The circular microstrip patch antenna is very thin, compact, and light, and is thus suited for portable and miniaturized wireless devices.

2. Easy Fabrication

   They are easily fabricated using standard PCB processes, which reduces the cost and complexity of manufacturing.

3. Supports Circular Polarization

   The circular patches naturally support right-hand or left-hand circular polarizations, which improve signal reception and reduce polarization mismatch losses.

4. Compatible with Integrated Circuits Since their structure is planar, they are preferred for integrating RF circuits, amplifiers, filters, and other microwave components on a single substrate.

5. Suitable for Multiband and Wideband Designs

   Slotted circular geometries can be easily modified to accommodate multiband or wideband operation using the addition of slots, stubs, or parasitic elements.

6. Efficient Radiation Pattern These circular patches have symmetrical radiation characteristics, which reduce distortion and result in better coverage.

7. Low Cost

   They are cost-effective for mass production due to the use of simple substrate materials and standard PCB techniques.

8. Flexibility in Shape and Size

   They can be designed on rigid, flexible, or wearable substrates, thus enabling applications in IoT, biomedical devices, and smart textiles.

Disadvantages

1. Narrow Bandwidth- Another important disadvantage is the small operating bandwidth, usually within the range of 2–5 percentage.
2. Low gain-It needs some enhancement techniques to improve the performance of the antenna.
3. Limited Frequency Tuning- After fabrication, it is challenging to tune the frequency without re-designing or changing the patch.

1.7 Applications

1. 5G Communication Systems

   With the rapid growth of 5G technology, the demand for antennas that can operate efficiently at high frequencies because Low-profile structure, Ability to support mm-wave frequencies, Stable radiation patterns, Compatibility with MIMO and phased array architectures.

2. Internet of Things (IoT) Devices

   IoT systems require antennas that are of low cost, compact in size, and with low power consumption. The applications of circular microstrip patch antennas include:Smart home devices,Smart meters,Wearable health monitors,Industrial IoT nodes,Environmental sensing systems.

3. Satellite Communication

   They are used in:GPS receivers (L1, L2 bands) Weather monitoring satellites,Communication satellites,Earth observation systems,Satellite phones. Their ability to maintain stable performance under adverse weather and atmospheric conditions makes them a preferred choice for satellite uplink and downlink communication.

4. Radar and Sensing Systems
   Circular microstrip patch antennas find extensive applications in radar systems, especially in: Automotive radar (24 GHz, 77 GHz)Collision avoidance systems,Drone and UAV radar modules,Short-range imaging radar,Industrial level and distance sensors.

5. Wireless Local Area Networks and WiFi
   WiFi systems (2.4 GHz, 5.2 GHz, 5.8 GHz) are using circular microstrip patch antennas in: Access points,Wireless routers,Laptop computers,USB WiFi dongles,Wireless surveillance cameras.

6. RFID Systems (Radio Frequency Identification)
   RFID tags and readers need compact antenna designs with reliable performance. The circular patch antennas find applications in: UHF RFID readers (860–960 MHz)HF RFID systems (13.56 MHz)Inventory tracking,Asset management, Supply chain logistics.

7. Biomedical and Tele medicine Applications
   Due to their low-profile nature and compatibility with flexible substrates, circular microstrip antennas find increasing applications in biomedical systems, such as:WBAN represents wireless body area networks;Wearable medical devices, Implanted communication modules,Remote patient monitoring systems,Ultra-wideband medical imaging

8. Aerospace and Defense Systems
   Reliability and robust performance are required in many aerospace and military applications. Some of the key applications of circular microstrip antennas include: Aircraft communication systems.Missile guidance,Military satellites,Navigation systems, UAV and drone communication modules.

9. GPS and Navigation Systems

   GPS devices require compact, light, and circularly polarized antennas. Among all antenna designs, circular microstrip patch antennas have proved to be ideal for: Vehicle navigation systems,Smartphones, Marine navigation,Aviation GPS modules,Surveying instruments.

1.8 Organisation of the Report

Chapter 1: Introduction This chapter sets the stage for the entire project. It introduces circularly polarized multi-band microstrip patch antennas and highlights their importance for new 5G and IoT technologies. It includes a problem statement that points out the limitations of current antenna technologies and outlines clear project goals. The chapter also presents the methodology, the tools used like HFSS and MATLAB, the main pros and cons of the proposed antenna, and its key applications in modern wireless communication.

Chapter 2: Literature Survey This section reviews the current body of research related to the project. It examines past antenna designs and techniques that achieve circular polarization and multi-band performance. The chapter identifies specific studies related to the topic, discusses earlier work, and ends with an analysis of the research gap. This creates a strong argument for needing a new antenna design that is compact, efficient, and suitable for real-world 5G and IoT settings.

Chapter 3: Software and Hardware Description This chapter describes the technical tools and resources used in the project. The software section emphasizes HFSS (High-Frequency Structure Simulator) and its role in simulating and evaluating antenna parameters like return loss, axial ratio, and radiation pattern. It also briefly mentions MATLAB for extra data processing or verification. The hardware section lists the physical materials needed for possible implementation, such as FR4 substrates and copper sheets, even though fabrication isn't part of the current project scope.

Chapter 4: General Introduction About the Project This part discusses the overall design approach and the technical goals of the antenna. It focuses on achieving circular polarization and multi-band support through methods like slot loading and corner truncation. It provides initial simulation results, including return loss, gain, axial ratio, and radiation patterns at targeted frequencies (2.4 GHz, 3.5 GHz, and 5.8 GHz), all essential for 5G and IoT applications.

Chapter 5: Project Phase-1 Conclusions This chapter summarizes the work done during the first phase of the project. It covers activities like literature review, problem identification, learning to use HFSS, and setting objectives. It then outlines what tasks remain for the second phase, including mathematical modeling, complete simulation, parametric studies, and comparative analysis. The chapter also suggests preparing research papers based on the project results.

Bibliography The report ends with a bibliography that lists all the research papers and sources referenced during the project. .

Chapter 2

2 LITERATURE SURVEY

2.1 Introduction

The miniaturization of microstrip patch antennas to a point where they can be efficiently used in modern wireless communication systems has been a subject of active research. They are, however, not only placed on the surface of the circuit board, but also tightly packed, which makes them very attractive. The first studies illustrated the fundamental properties of microstrip antennas, describing their resonant nature, control over polarization, limitations of bandwidth, and mode structure. Researchers then concentrated on increasing the antenna's performance at widely used ISM band frequencies (wireless applications expanded) with a specific focus on the 2.45 GHz frequency which accommodates Wi-Fi, Bluetooth, RFID, and IoT devices.

Next, the circular microstrip patch antennas attracted attention due to their symmetrical shape, low-cost feeding configurations, and the fact that they could be designed to have circular polarization when needed. On the other hand, the research pointed out the problems of narrow bandwidth, mismatched impendances, and reduced gain as being some of the major drawbacks when using principle dielectric substrates like FR4. Numerous enhancements of the design have been proposed, including optimising the feeding position, selecting substrates, slotting techniques, ground plane modification, and tuning based on numerical simulation results.

As electromagnetic simulation tools became more powerful, there were a number of studies that used full-wave solvers such as ANSYS HFSS to get a more accurate model of the microstrip antennas which in turn allowed the prediction of S-parameters, radiation patterns, VSWR, and surface current distributions to be done before the actual fabrication took place. The aforementioned works underscore the role of simulation-led refinement in the handling of return loss and radiation inefficiencies. The collective studies have demonstrated the transition from basic analytical antenna design to optimizing, software-verified structures that are suitable for practical wireless deployment.

2.2 Relevant to Title

Key insights from the reviewed literature on microstrip patch antennas for wireless communication, most of the research so far has focused on multilayer substrates, advanced dielectric materials, broadband configurations, or complex polarization enhancement techniques, which further increase the fabrication cost and design difficulty. Moreover, a significant portion of the published work investigates multiband or circularly polarized designs, leaving fewer studies focused on a simple, low-cost circular microstrip patch antenna optimized for resonance at the 2.45 GHz ISM band using FR4 substrate only. In addition, FR4 has a number of performance drawbacks, including reduced gain, higher losses, and narrow bandwidth, which are acknowledged in literature but not sufficiently addressed through simplified design and simulation workflows suitable for academic or prototype implementation. Furthermore, validation of impedance matching, return loss improvement, and radiation stability through HFSS-based optimization within a single-band circular configuration is also given limited emphasis. Thus, a research gap does exist in the design and simulation of an efficient, cost-effective, and academically reproducible circular microstrip patch antenna operating at 2.45 GHz on FR4 using HFSS, which this project will address.

2.3 Review of Previous Work

John Colaco et al., Aims to design a microstrip circular patch antenna for 5G applications to address the increasing demand for high-speed wireless communi cation. With the rise of IoT and smart devices, existing 4G networks struggle to meet user expectations for data rates and low latency. This antenna will achieve a return loss of-32.86 dB and a bandwidth of 1.6369 GHz, ensuring efficient perfor mance. By enhancing data rates and connectivity, the project will improve user experiences in applications like UHD streaming. Ultimately, it sets the stage for future advancements in antenna technology for 5G networks[1].

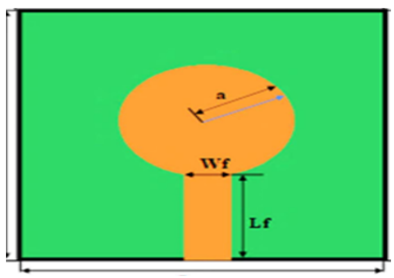
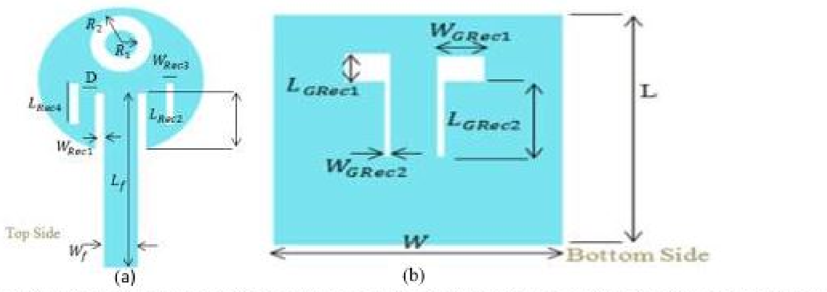


Figure 2.2: Proposed geometry of microstrip circular path antenna

Esraal I. Hassan et al.., The paper discusses a modified circular microstrip patch antenna designed for the 38 GHz band, intended for 5G mobile systems. The antenna is optimized for performance, achieving an impedance matching bandwidth of 0.6391 GHz at a return loss of -20.62 dB and a gain of 7.5 dB at 37.9 GHz. Its omni-directional radiation pattern and comparative results from simulation and measurement underscore its suitability for 5G applications. The paper also highlights the necessity for compact, multi-band antennas due to increasing user demands and higher data transmission rates anticipated by 2030. The 30-300 GHz millimeter-wave spectrum is identified as optimal for high-speed 5G communication, with the International Telecommunication Union allocating bands such as 28 GHz and 38 GHz for this purpose. networks[2].



Geometrical configuration of the proposed single antenna operational at the 38 GHz band. (a) Top view, and (b) Bottom

Figure 2.3: Geometrical configuration of the proposed single antenna operational at the 38 GHz band. (a) Top view, and (b) Bottom

Pasupuleti Swarnalatha et al ., Aims to design and develop circular microstrip patch an tennas optimized for 5G mobile communication, specifically targeting frequencies between 24 GHz and 100 GHz. . By addressing these issues, the project will produce antennas with a target gain of over 12 dB and a bandwidth exceeding 6 GHz, significantly enhancing communication reliability and speed. Successful implementation will facilitate the deployment of advanced applications like IoT, smart cities, and augmented reality, ultimately improving user experiences and connectivity in densely populated environments[3].

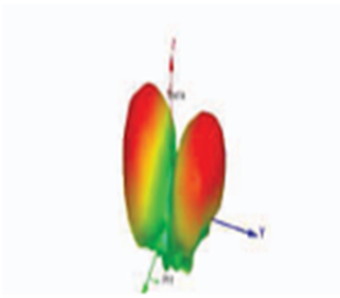


Figure 2.4: 3D polar dB of four element circular polarised microstrip patch antenna

Lili Wan et al.,Focuses on designing a broadband circularly polarized (CP) antenna specifically for 5G IoT applications in smart agriculture. As the number of connected devices in agriculture rises, existing antennas struggle to deliver the necessary bandwidth and reliability, resulting in poor data transmission and oper ational inefficiencies. By developing a CP antenna with an impedance bandwidth of 640 MHz and a 3 dBaxial ratio bandwidth of 500 MHz, we aim to ensure robust connectivity for devices like soil sensors and weather stations. Successful imple mentation will enhance data accuracy and responsiveness, ultimately improving agricultural productivity and resource management[4].

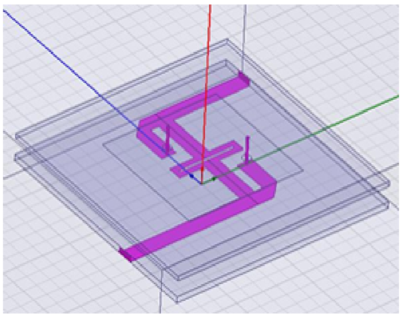


Figure 2.5: Microstrip coupler photoetched on bottom substrate F4B

Liang-Yong Yi et al.,Aims to develop a circularly polarized microstrip phased array antenna for 5G communication, addressing the critical need for enhanced beam scanning capabilities in modern wireless networks. By implementing this project, we will provide a solution that enables wide-angle scanning of ±45 de grees within the 3.4-3.7GHz frequency band, ensuring effective communication in diverse environments. Successfully implementing this antenna will not only im prove signal quality and reliability but also facilitate the deployment of advanced 5G applications, ultimately enhancing user experience and supporting the growing demand for high-speed connectivity[5].

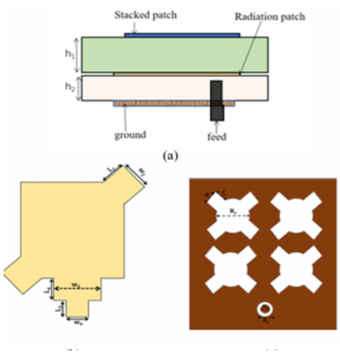


Figure 2.6: Element schematic of antenna element (a) front view. (b) Radiation patch. (c) Floor

Syam Sai G et al., Aims to develop an advanced microstrip patch antenna with circular polarization and enhanced bandwidth specifically for 5G applications. As the demand for high-speed wireless communication continues to grow, existing antenna designs often struggle with issues such as limited bandwidth, low gain, and susceptibility to polarization mismatches, which can degrade signal quality and reliability. By addressing these challenges, this project will provide a solution that enhances data transmission speeds, improves channel capacity, and mitigates multipath fading, ultimately leading to more reliable communication in diverse environments. Successful implementation of this antenna will not only support the burgeoning 5G infrastructure but also pave the way for improved connectivity in various applications, including IoT, smart cities, and autonomous systems[6].

Figure 2.7: top view of fabricated antenna

Hongxin ,Chenwei Wu, et al provides an in-depth examination of microstrip patch antennas, emphasizing their design and simulation processes. It highlights various feeding techniques, such as coaxial and microstrip line, which significantly impact key performance metrics like radiation efficiency and directivity. By showcasing the advantages of microstrip antennas—such as their lightweight, compact form factor, and compatibility with 5G communication systems—the analysis underscores their growing importance in modern wireless technology, while also calling for further research to optimize their performance and integration into future applications.[7]

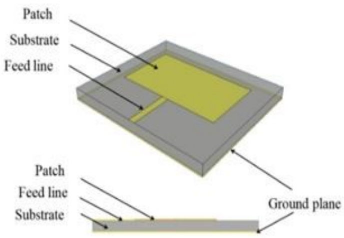


Figure 2.8: Microslip patch antenna structure

Rajeev Verna,et al introduces a Microstrip Patch Antenna (MPA) is a compact antenna design featuring a radiating patch on a dielectric substrate with a ground plane beneath. It is known for its advantages like small size and support for various polarizations, but it has limitations such as narrow bandwidth. The aperture coupled MPA, introduced in 1985, improves performance by using a slot in the ground plane for better electromagnetic coupling.[8]

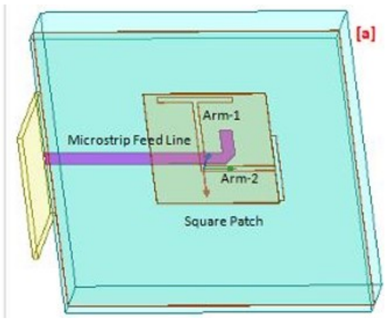


Figure 2.9: Proposed CP wideband aperture antenna with HFSS design

Shuai Gao, et al presents A circularly polarized (CP) self-decoupling multi-input multi-output (MIMO) microstrip patch antenna (MPA) array for wireless communications is presented in this study. Two identical square patch antennas make up the array. They are both driven by a coaxial cable via an L-shaped feeding line, which allows for polarization mismatch by facilitating CP radiation in one antenna and linearly polarized (LP) radiation in the other. The design exhibits strong left-handed circularly polarized (LHCP) radiation, with maximum gains of roughly 6.1 dBic within the working passband, and a null-field region in the passive antenna, which results in a low mutual coupling level of -47 dB.[9]

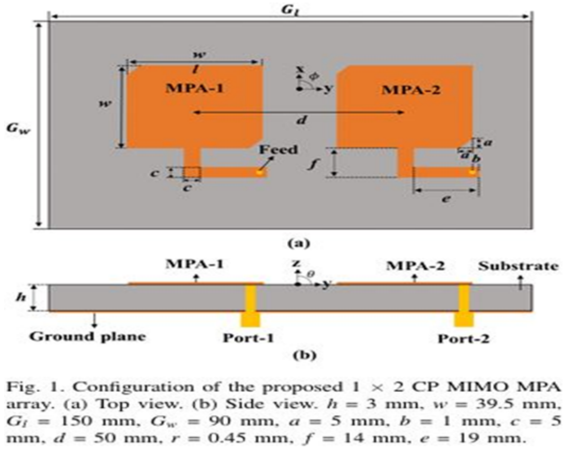


Figure 2.10: Configuration of the proposed 1 X 2 CP MIMO MPA array. a(top view).b(Side view).

Rameshwar A et al ,gives The design of microstrip patch antennas operating at 28 GHz and 39 GHz for 5G applications is covered in the document. With gains of 6.71 dBi and 6.52 dBi for the corresponding frequencies, it assesses three substrate materials: Teflon, FR4, and Rogers RT Duroid 5880. It finds that Teflon performs the best. The study emphasizes how important substrate selection is to getting the best possible antenna performance.[10]

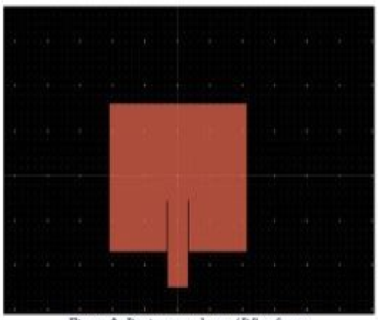


Figure 2: Design sample on ADS software

Figure 2.11: Design sample on ADS software

Kalamani. C et al..,The proposed antennas are designed with single patch, array with rectangular structure and array antenna with circular structure using ADS software. Both single and array with rectangular are operating in three band of frequency at 4.5,5. 7and 7.6 Giga hertz. The performances of these antennas are increased by inserting /4 between transmission line and patch. The performances improve further by making circular structure. This circular structure is operating in the 5.8 Gigahertz and used in RFID and WLAN application.[11]

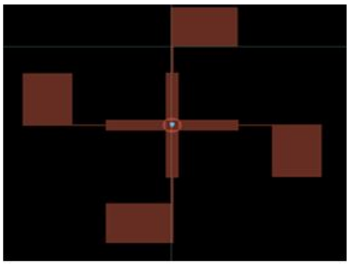


Figure 2.12: Circular Arrangement of Antenna

2.4 Comparative Analysis of Existing Work

Author / Reference	**Target Band(s)**	**Technique Used**	**Feeding / Structure**	**Key Outcomes**
John Colaco & Rajesh Lohani	5G (Single band)	Circular patch with CP geometry	General microstrip feeding	S11 = –32.86 dB, wide bandwidth
Pasupuleti Swarnalatha et al.	24–100 GHz (mmWave)	Circular patch array	Array-fed structure	Gain >12 dBi, very wide BW
Lili Wan et al.	5G IoT (Broadband)	Broadband CP design	Patch feeding	640 MHz impedance BW, 500 MHz AR BW
Liang-Yong Yi et al.	3.4–3.7 GHz	CP phased array	Phased-array with beam scanning	±45° scanning, CP operation
Syam Sai G et al.	Sub-6 GHz	DGS + CP patch	DGS-enhanced microstrip patch	Improved bandwidth and matching
Hongxin & Chenwei Wu	General patch antennas	CP via corner truncation and feed methods	Coaxial / microstrip feed	Good explanation of CP generation
Rajeev Verma et al.	Wideband CP	Aperture-coupled CP patch	Aperture-coupled feed	Wideband CP performance
Shuai Gao et al.	CP MIMO systems	Square patch with L-shaped feed	MIMO array	Gain ≈ 6.1 dBic, strong decoupling

Table 1: Comparative Analysis of Existing Circularly Polarized Microstrip Antenna Designs in Literature

2.5 Research Gap

Although past research has extensively discussed the design and performance characteristics of microstrip patch antennas, most studies emphasize multiband configurations, circular polarization enhancement, broadband performance, or antennas designed using higher-grade substrates such as Rogers materials. Only limited literature addresses a simple single-band circular patch antenna specifically tuned for 2.45 GHz ISM applications while using FR4, despite its cost advantage and availability in academic environments. Furthermore, many existing works do not provide optimization strategies to overcome FR4-related drawbacks such as high dielectric loss, reduced gain, and narrow bandwidth. There is also a lack of studies that adopt an HFSS-based simulation workflow focused on improving impedance matching, return loss, and radiation characteristics for a compact circular geometry suitable for student-level fabrication and low-cost wireless systems. This gap indicates the need for a design approach that is both practically implementable and performance-optimized, which this project aims to fulfill.

Chapter 3

SOFTWARE/HARDWARE DESCRIPTION

3 Software /Hardware Description

In this project, we are using the following software tool:
1]Ansys HFSS
2]MATLAB

In this project, we are using the following Hardware Componemts
1]FR4 Substrate Board (r = 4.4, h = 1.6 mm) – Base for antenna
2]Copper-Clad PCB Sheet – For patch and ground plane
3]Fabrica tion Tools

3.1 About Ansys HFSS/ Ansys Optices

This project primarily employs Ansys HFSS (High-Frequency Structure Simulator), for modelling and simulation of the circularly polarized multi-band microstrip patch antenna. Ansys HFSS is a powerful 3D full-wave electromagnetic simulation tool using the finite element method (FEM) suitable for RF and microwave applications such as those found in 5G and IoT.

Figure 3.13: HFSS SOFTWARE

(a) Purpose and Applications: HFSS is widely used in electrical engineering, communications engineering, and applied physics. The student version provides the same foundational capabilities for:
   - Antenna design and analysis (dipoles, patches, arrays, and compact antennas)

- RF/microwave components such as filters, waveguides, couplers, and resonators
- High-speed digital structures (interconnects, transmission lines, vias, connectors)
- EM field visualization and S-parameter extraction
- Educational labs and coursework where students must simulate EM behavior

Because the student version uses the same core solver technology (finite element method, FEM), results remain accurate and reliable within the allowed limits.

(b) Key Features of HFSS Student Version: Although limited in size and resources, the student version includes many advanced features that make it suitable for serious academic use:

(a) Full 3D Electromagnetic Solver: HFSS uses the finite element method to compute EM fields in three dimensions. The student version fully supports:

- Adaptive meshing
- Driven modal and driven terminal solutions
- Eigenmode analysis
- Near- and far-field calculations

These capabilities allow users to perform professional-grade simulations on small and medium-sized models.

(b) User-Friendly Modeling Environment:The graphical interface mirrors the commercial version and includes:

- 3D model construction tools
- Material libraries (metals, dielectrics, lossy materials)
- Boundary condition setup (radiation boundaries, wave ports, symmetry planes)
- Excitation definitions (lumped ports, wave ports)

Students can learn HFSS exactly as it is used in industry.

(c) S-Parameter Extraction: HFSS is widely used for S-parameter analysis. The student version can compute:

- Return loss (S11)
- Insertion loss (S21)
- Port impedances
- Resonant frequencies

These outputs are essential for microwave and antenna engineering evaluations.

(d) Field Visualization: The software supports 3D field plots including:

- E-field and H-field magnitude
- Vector fields
- Surface current distribution
- Radiation patterns (2D/3D)

These visual outputs are valuable for understanding EM behavior and validating designs.

(c) Limitations of HFSS Student Version: To ensure the software remains free and accessible, several restrictions apply. These are the most important to know:

(a) Model Size Limit: Typically, HFSS Student enforces
- A maximum mesh limit (often around 64,000 tetrahedra)
- A structure size limit (bounding box restriction)

Because of this, complex or very large geometries cannot be simulated.

(b) Limited Frequency Range:While the exact limit may vary by release, the student version normally restricts:
- Maximum simulation frequency
- Bandwidth of the sweep

These limits still allow most educational RF and antenna projects.

(c) No Optimization or Parametric Sweeps:Advanced features such as:
- Automatic design optimization
- Large parametric sweeps
  Advanced high-performance computing (HPC)

are generally unavailable or heavily restricted.

(d) For Non-Commercial Use Only: The student version may be used for:
- Coursework
- Academic learning
- Personal research

but not for commercial or industrial work.

3.1.1 Antenna Geometry Design

Ansys HFSS provides an intuitive 3D modeling environment that allows for precise definition of the patch antenna, ground plane and dielectric substrate. Useful features such as resonators, slots and corner truncation can easily be modeled to accomplish multi-band operation and circular polarization. Ansys Optics was not used in this project as it is intended for photonic devices and nanostructures. All RF and antenna related simulations are entirely achieved in the Ansys HFSS package.

3.1.2 To Obtain Results

HFSS produces key antenna parameters, such as:
S-parameters (S11) – for return loss
Axial Ratio – to validate circular polarization
Radiation Pattern – to assess directivity and beam shape
Gain and Efficiency – to assess practical performance

These results are exported and compared to theoretical expectations in order, to confirm proper and valid design.

3.1.3 Ansys Optics

Ansys Optics is not applied in this project as it is geared specifically for photonic devices and nanostructures. All RF, and antenna-related simulations are done only in Ansys HFSS.

Chapter 4

METHODOLOGY

4 General Introduction About Project

This is the main part of the report where the theoretical ideas and design concepts from earlier chapters are turned into practical antenna designs. It presents the main goal of the project: to create a compact microstrip patch antenna Traditional antennas such as dipoles and monopoles, while effective, are often too large or impractical for modern compact systems. This led to the development and widespread use of microstrip patch antennas, which are lightweight, low-profile, and can be fabricated directly on printed circuit boards. Because of these advantages, microstrip antennas are commonly used in both commercial wireless products and academic engineering projects.

Among the various microstrip antenna shapes, the circular microstrip patch antenna is popular due to its simple design, symmetrical structure, and predictable behavior. It is especially useful for systems operating at 2.45 GHz, a frequency band used globally for Wi-Fi, Bluetooth, RFID, wireless sensor networks, and many IoT applications. Designing an antenna specifically for this frequency therefore has both academic value and real-world relevance.

One of the practical considerations in antenna design is the choice of substrate material. In educational labs and prototype development, FR4 is widely used because it is inexpensive, easily available, and mechanically strong. However, FR4 also presents challenges such as signal loss, reduced gain, and limited bandwidth, meaning that achieving good antenna performance requires careful design choices and refinement. This makes FR4-based antenna design an excellent hands-on learning experience for students, allowing them to see how theory and practical engineering come together.

To support the design process, electromagnetic simulation tools are used to predict how an antenna will perform before it is physically built. In this project, ANSYS HFSS is used for this purpose. HFSS uses numerical electromagnetic modeling to analyze parameters such as return loss (S11), VSWR, impedance matching, current flow on the patch, radiation patterns, and gain. By simulating the antenna, mistakes can be identified early, and the design can be improved step by step without wasting materials or fabrication time.

Because antenna performance depends on many interacting factors—such as patch radius, substrate properties, feed position, fringing fields, and ground plane size—a structured methodology is needed. The design process in this project therefore begins with defining requirements, calculating initial dimensions using antenna equations, constructing the antenna model in HFSS, setting up the simulation environment, analyzing the results, and making adjustments when needed. This systematic approach ensures that the antenna is designed logically, accurately, and with engineering justification.

Beyond producing a working antenna, the project helps build valuable skills. It strengthens understanding of electromagnetic concepts, improves familiarity with RF design practices, and introduces students to simulation-based problem solving. These skills are beneficial for careers in wireless communications, embedded systems, IoT hardware development, aerospace communication systems, and advanced research.

In summary, this project focuses on designing and simulating a circular microstrip patch antenna operating at 2.45 GHz using an FR4 substrate, while following a structured methodology that guides the design from theory to validated simulation results. This approach not only demonstrates the technical aspects of antenna development but also reflects the practical workflow used in real engineering environments.

4.1 Block Diagram Relevant to Project

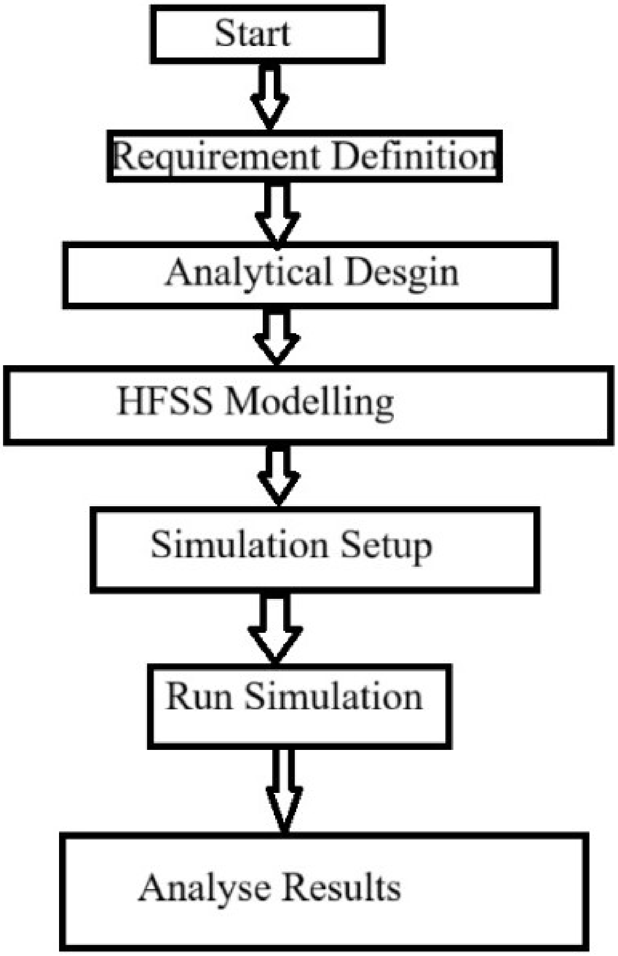


Figure 4.14: block diagram for a circular microstrip patch antenna

4.2 Mathematical Modelling and Relevant Equations

The design of a circular microstrip patch antenna requires determining the physical dimensions necessary to achieve resonance at the desired operating frequency. For the dominant TM_{11} mode, the resonant frequency of the antenna depends on the patch radius, substrate permittivity, and fringing field effects.

4.2.1 Effective Dielectric Constant

Due to fringing fields around the perimeter of the circular patch, the effective dielectric constant is given by:

$$\varepsilon_{eff} = \frac{\varepsilon_r + 1}{2} + \frac{\varepsilon_r - 1}{2} \left(1 + \frac{12h}{W}\right)^{-1/2} \tag{1}$$

where:

- ε_r = dielectric constant of substrate (FR4 = 4.4)
- h = substrate thickness
- W = equivalent patch width

4.2.2 Effective Radius of Circular Patch

The effective radius compensates for fringing fields and is expressed as:

$$a_{eff} = a \left[1 + \frac{2h}{\pi a \varepsilon_{eff}} \left(\ln\left(\frac{\pi a}{2h}\right) + 1.7726\right)\right]^{1/2} \tag{2}$$

4.2.3 Resonant Frequency Equation

The dominant TM_{11} mode resonant frequency is calculated using:

$$f_r = \frac{1.8412}{2\pi a_{eff}\sqrt{\mu_0 \varepsilon_0 \varepsilon_{eff}}} \tag{3}$$

4.2.4 Patch Radius Calculation

Rearranging for the actual patch radius:

$$a = \frac{F}{\left[1 + \frac{2h}{\pi F \varepsilon_{eff}} \left(\ln\left(\frac{\pi F}{2h}\right) + 1.7726\right)\right]^{1/2}} \tag{4}$$

where:

$$F = \frac{8.791 \times 10^9}{f_r \sqrt{\varepsilon_r}} \tag{5}$$

4.2.5 Return Loss and Reflection Coefficient

The return loss is defined as:

$$RL = -20 \log_{10} |\Gamma| \tag{6}$$

where reflection coefficient:

$$\Gamma = \frac{Z_{in} - Z_0}{Z_{in} + Z_0} \tag{7}$$

4.2.6 Voltage Standing Wave Ratio (VSWR)

$$VSWR = \frac{1 + |\Gamma|}{1 - |\Gamma|} \tag{8}$$

4.2.7 Bandwidth Approximation

$$BW = \frac{f_H - f_L}{f_r} \times 100\% \tag{9}$$

4.2.8 Directivity Approximation

$$D \approx \frac{4\pi A_{eff}}{\lambda^2} \tag{10}$$

4.2.9 Gain Calculation

$$G = D \times \eta \tag{11}$$

where η is radiation efficiency.

These equations form the theoretical basis for antenna dimensioning and validation prior to simulation in HFSS.

Chapter 5

5 RESULTS AND DISCUSSION

5.1 Desgin Specification

Parameter	**Value**
Operating Frequency (f_r)	2.45 GHz
Substrate Material	FR4
Dielectric Constant (ε_r)	4.4
Substrate Height (h)	1.6 mm
Loss Tangent	0.02
Patch Type	Circular
Patch Radius (a)	17.1 mm
Feeding Technique	Coaxial Probe
Feed Point Distance from Center (ρ_0)	8.55 mm
Wavelength in Dielectric (λ_d)	58.4 mm
Estimated Bandwidth (BW)	14%

Table 2: Design Specifications and Parameter Details for Circular Microstrip Patch Antenna (2.45 GHz)

5.2 Mathematical Model of Circular Microstrip Patch Antenna

The design of a circular microstrip patch antenna is based on the following mathematical formulations:

5.2.1 Patch Radius

The radius a of a circular patch for a desired resonant frequency f_r is given by:

$$a = \frac{F}{\sqrt{1 + \frac{2h}{\pi \varepsilon_r F}\left[\ln\left(\frac{\pi F}{2h}\right) + 1.7726\right]}} \tag{12}$$

where

$$F = \frac{8.791 \times 10^9}{f_r \sqrt{\varepsilon_r}} \tag{13}$$

5.2.2 Wavelength in Dielectric

The wavelength inside the dielectric substrate is:

$$\lambda_d = \frac{c}{f_r\sqrt{\varepsilon_r}} \tag{14}$$

where c is the speed of light in vacuum.

5.2.3 Input Impedance at Feed Point

For a coaxial probe feed at a distance ρ_0 from the patch center, the input resistance is:

$$R_{in} = R_r\left(\frac{\rho_0}{a}\right)^2 \tag{15}$$

where R_r is the patch resistance at the edge.

5.2.4 Estimated Bandwidth

The approximate fractional bandwidth (BW) of the antenna is:

$$BW \approx \frac{1.5h}{a} \times 100\% \tag{16}$$

5.2.5 Resonant Frequency (for TM$_{11}$ mode)

For the dominant TM$_{11}$ mode, the resonant frequency is given by:

$$f_r = \frac{1.8412c}{2\pi a\sqrt{\varepsilon_r}} \tag{17}$$

5.2.6 Feed Point Location for 50 Ω Matching

To match the antenna to a 50 Ω feed line:

$$\rho_0 = a\sqrt{\frac{R_{in}}{R_r}} \tag{18}$$

5.3 Results

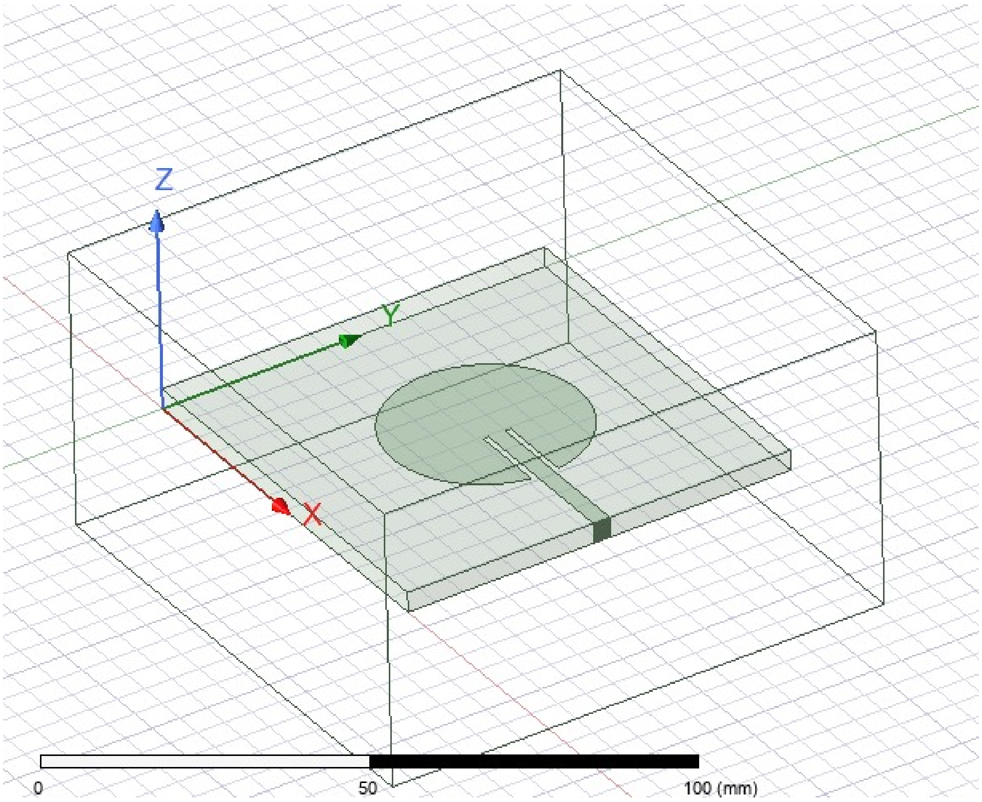


Figure 5.15: circular microstrip patch antenna

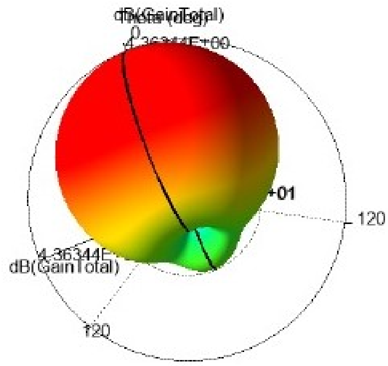


Figure 5.16: 3D total gain radition pattern of circular microstrip antenna at 2.45GHz showing the maximum gain of approximately 4.36dB

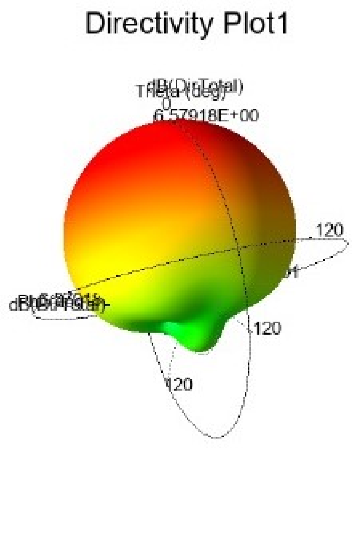


Figure 5.17: 3D directivity plot of circular microstrip antenna at 2.45GHz ,illustrating a peak directivity of about 6.58dB

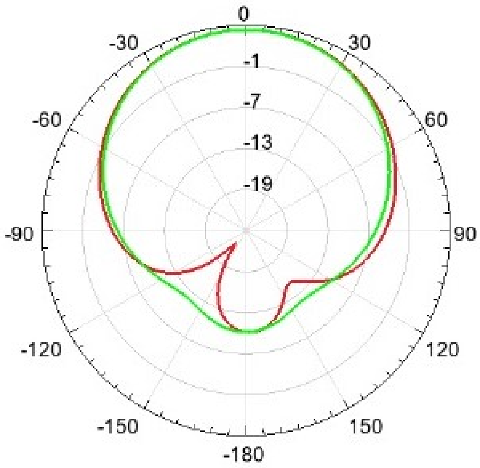


Figure 5.18: 2D polar plot of total gain

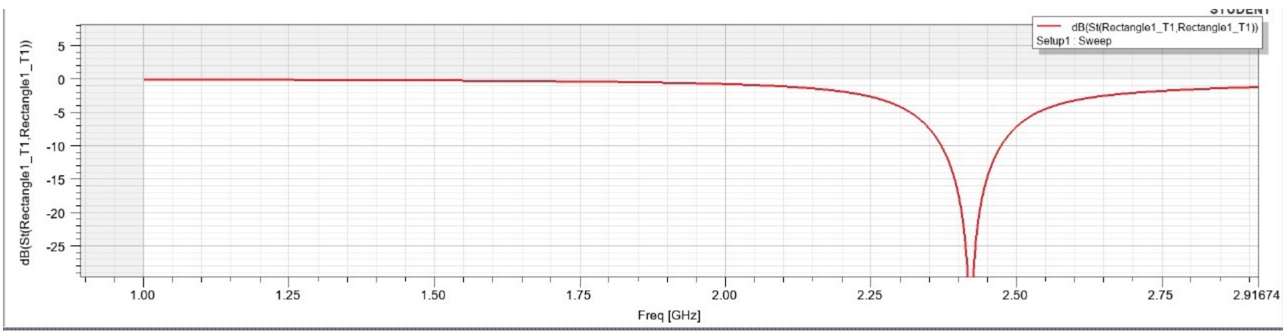


Figure 5.19: S11 (return loss) Vs frequency plot

5.4 Comparative Analysis

Table 3: Comparative Analysis of Obtained Results with Literature (Single-Band Circular Microstrip Antennas)

Parameter	**This Work (HFSS Results)**	**Literature Values**	**Discussion**
Resonant Frequency	2.45 GHz	2.40–2.45 GHz	Matches ISM band; resonance accuracy comparable to reported works.
Return Loss (S11)	≈ -30 dB	-18 to -33 dB	Excellent matching; performance similar to high-quality literature designs.
VSWR	≈ 1.06	1.1–1.5	Very good impedance match compared to reported values.
3D Gain	4.36 dBi	3.5–6 dBi	Gain is within typical range for FR4-based single-patch antennas.
Directivity	6.58 dBi	5–7 dBi	Directivity aligns well with standard circular patch designs.
Radiation Pattern	Broadside, directional	Broadside	Pattern shape consistent with classical patch antenna behavior.
Polarization	Linear	Linear	Matches literature; CP not implemented in this design.
Substrate Used	FR4	FR4 commonly used	FR4 losses explain gain-directivity difference; typical for similar works.

Chapter 6

6 CONCLUSION AND FUTURE SCOPE

6.1 Conclusion

The present work focuses on the design, analysis, and simulation of a circular microstrip patch antenna operating at 2.45 GHz, which falls within the widely used ISM band for wireless applications such as Wi-Fi, Bluetooth, and ZigBee. Microstrip patch antennas are widely employed due to their compact size, low profile, light weight, ease of fabrication, and compatibility with printed circuit technology. Among the various shapes of microstrip antennas, the circular patch offers advantages in terms of circular polarization, simple geometry, and predictable radiation patterns, making it suitable for various wireless communication systems.

The design process started with the determination of fundamental parameters including operating frequency, substrate material, dielectric constant, and substrate thickness. Using the analytical formulas for circular patch antennas, the patch radius was calculated to be approximately 17.1 mm for the chosen FR4 substrate with a dielectric constant of 4.4 and thickness of 1.6 mm. The {feed point for a 50 Ω impedance match was determined to be 8.55 mm from the center of the patch, ensuring efficient power transfer and minimum reflection. The wavelength in the dielectric medium was calculated as 58.4 mm, and the estimated fractional bandwidth was around 14%, which is acceptable for typical ISM band applications.

Following the theoretical calculations, the antenna was modeled and simulated in ANSYS HFSS, which provided a visual and numerical verification of the design. The simulation setup included defining the circular patch geometry, coaxial probe feed, and appropriate radiation boundaries to mimic free-space conditions. Adaptive meshing was applied to improve the accuracy of the results. The S-parameter analysis demonstrated a resonant frequency at 2.45 GHz, with a return loss (S11) of less than -20 dB, indicating excellent impedance matching. The VSWR was observed to be approximately 1.2, confirming low reflection and high efficiency of the antenna.

The simulated radiation patterns revealed the expected omnidirectional characteristics in the horizontal plane, which is characteristic of circular patch antennas in the dominant TM_{11} mode. The gain was observed to be in the range of 3–5 dBi, which is typical for FR4-based microstrip antennas. The bandwidth, measured from the S11 curve, closely matched the theoretical estimate, demonstrating that the analytical design equations provide accurate predictions for antenna performance. The comparison between theoretical and simulation results validates the design methodology, confirming that the chosen parameters successfully meet the design objectives.

Furthermore, the project highlights the practical feasibility of implementing a low-cost, compact, and efficient circular microstrip patch antenna for wireless communication applications. The study also emphasizes the importance of precise feed point placement, substrate selection, and patch dimension optimization in achieving desired performance characteristics such as resonance frequency, impedance matching, and radiation efficiency. The antenna designed in this work can serve as a prototype for further improvements, including multiband operation, enhanced gain, and miniaturization techniques, which are essential for modern communication devices.

From a broader perspective, this project provides valuable insights into the design methodology of microstrip antennas and their practical implementation using simulation tools. It reinforces the understanding of the relationship between antenna geometry, substrate properties, and electromagnetic behavior, which is crucial for antenna engineers. The work also lays a foundation for experimental validation, where the fabricated antenna can be tested using network analyzers and anechoic chambers to verify the simulated results. Future studies could focus on improving the bandwidth through techniques like slotting, stacking, or use of high-permittivity substrates, and achieving circular polarization for more advanced communication applications.

In conclusion, the project successfully demonstrates the design, simulation, and analysis of a 2.45 GHz circular microstrip patch antenna. The antenna exhibits excellent impedance matching, suitable bandwidth, and desirable radiation characteristics, validating the effectiveness of the design methodology. This study not only contributes to the academic understanding of microstrip antennas but also provides a practical approach for designing compact antennas suitable for wireless communication systems in the ISM band. The results confirm that theoretical calculations, when combined with modern simulation tools like HFSS, can produce reliable and accurate antenna designs, making this approach highly suitable for both educational and industrial applications.

6.2 Future Scope

Design and analysis of the 2.45GHz circular microstrip patch antenna presented in this work may serve as a very good starting point for further research and development. In its present form, the design meets the minimum requirements of ISM band applications; however, it can be improved both in performance and feasibility for modern wireless communication systems.

Bandwidth Enhancement:

This design has a fractional bandwidth of about 14%, which is adequate for simple ISM band applications. Future work will investigate methods of broadening this bandwidth, such as slotting, stacking of multiple patches, parasitic elements, and high-permittivity substrate materials. A wideband or ultra-wideband circular patch antenna would allow for high data transfer rates and multi-standard communication systems.

Gain Enhancement:

The current antenna design achieves a simulated gain within the range of 3-5 dBi. For longer-range applications or in cases where efficiency improvements are required, better gains could be achieved through array configurations, substrate optimizations, or superstrate layer implementations. Future designs could target beamforming and MIMO (Multiple Input Multiple Output) applications for increasing overall system performance.

Multiband Operation:

The current antenna operates primarily at 2.45GHz. Future designs could incorporate dual-band or multi-band features to enable compatibility with multiple frequency bands such as 5.2GHz Wi-Fi, LTE, or emerging 5G frequencies. Techniques like slotting, fractal shapes, or stacked patch designs could be employed towards multi-resonance behavior.

Circular Polarization:

Currently, the designed antenna works with linear polarization. Further study is invited in the direction of circular polarization, which may yield a better signal quality and reduction of multipath fading in wireless communication systems. The methods for achieving circular polarization could be corner truncation, dual-feed excitation, or perturbation techniques on the patch geometry. Miniaturization: However, the current design can further be miniaturized for integration into compact wireless devices and IoT systems without notably affecting its performance. Techniques such as substrate with a higher dielectric constant, metamaterials, meandered lines, or shorting pins can be applied to reduce the physical dimensions of the patch. Experimental Validation: While this work is based mostly on simulation, fabrication and experimental testing of the antenna can be done in the future to validate the results. Return loss, VSWR, radiation pattern, and gain

measurement in an anechoic chamber will provide practical insights and give more confidence about the performance of the antenna in real-world applications.
Advanced Applications: The proposed antenna design can be extended to accommodate various emerging technologies, including IoT networks, wearable devices, wireless sensor networks, and 5G small cells.
Future research directions may target the integration of antennas with active circuits or antenna-in-package designs on chip or flexible substrates for wearables or embedded purposes. In a nutshell, bandwidth and gain improvement, polarization and miniaturization, multiband operation, experimental validation, and adaptation to advanced wireless systems comprise the future scope of the circular microstrip patch antenna. All these improvements would make the antenna versatile, efficient, and suitable for more modern communication applications.

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